

# Analytical Framework for Risk Estimation and Independent Population Monitoring

Ecuadorian national registry · Unit of analysis: hospital admission

## ARI admission registry (2015–2023)

576,195 admissions | 41,048 in-hospital deaths

## Historical non-ARI records

9,463,108 discharges | Preceding 12 months  
18 facility-profile features (7 service statistics + 11 diagnostic proportions)

Individual features + historical facility profiles

Shared ARI cohort; independently defined references

## A Prediction branch: HiCARE risk estimation

### Base risk learning

33 base features + 18 facility-profile features  
LightGBM + XGBoost  
Nested predictions for second-layer fitting

### Stratified calibration

Interaction stacking

Residual correction

Marginal projection

$$p_i^H = \sigma\left(\beta_0 + \beta_1 z_i^B + \sum_k \gamma_k g_k(i)\right)$$

$z_i^B$ : residual-corrected score;  $g_k(i)$ : category indicator

### Predictive evaluation

5-fold cross-validation

| Model  | AUROC  |
|--------|--------|
| LR-7   | 0.8591 |
| EW     | 0.9028 |
| HiCARE | 0.9037 |

Weighted mean  $|\log(O/E)|$  across  
42 marginal groups

**EW 0.0219 → HiCARE 0.0054**

### Monthly state updating

$$\text{logit}(p_{i,t}) = \varphi_i^T \theta_t$$

$$\theta_t = \theta_{t-1} + \varepsilon_t$$

Update intercept, slope, COVID-19 and child offsets  
Predict first; update after outcomes become available

### Temporal holdout (2022–2023)

HiCARE static → monthly updating

**ICI 0.0196 → 0.0052**

EW with the same update: ICI 0.0051

## B Monitoring branch: task-specific independent references

### Regional and facility comparisons

Independent LightGBM case-mix model

$$O_g = \sum_{i \in g} y_i, \quad E_g = \sum_{i \in g} q_i$$

$$SMR_g = O_g / E_g$$

$q_i$ : independent reference probability  
Empirical Bayes shrinkage

**Adjusted mortality ratios across 24 provinces: 0.61–1.56**

### Non-COVID comparisons across periods

Fixed risk relationship fitted in 2015–2019  
Assess mortality departures in later periods

### Paediatric demand monitoring

Pre-pandemic seasonal count reference  
2023 observed / expected admissions

**1.32 (95% prediction interval 1.23–1.43)**

Fixed seasonal baseline

### Identify populations, regions and periods for review

Inform local service preparedness